pressure. Einstein's laws (Figure 6.7) insist that when any star is smaller than its critical circumference, its gravitational squeeze will be so strong that no internal pressure can possibly counterbalance it. The star has no choice but to implode.

2. Newton's laws (Figure 6.8) predict that photons emitted from the star's surface at first will fly out to larger circumferences, even in some cases to circumferences larger than critical, and then will be pulled back in. Einstein's laws (Figure 6.7) demand that any photon emitted from inside the critical circumference move always toward smaller and smaller circumferences. The only reason that such a photon can escape the star's surface is that the star itself is shrinking faster than the outward-directed photon moves inward (Figure 6.7).

Although Finkelstein's insight and the bomb code simulations fully convinced Wheeler that the implosion of a massive star must produce a black hole, the fate of the imploding stellar matter continued to disturb him in the 1960s, just as it had disturbed him in Brussels in his 1958 confrontation with Oppenheimer. General relativity insisted that the star's matter will be crunched out of existence in the singularity at the hole's center (Chapter 13), but such a prediction seemed physically unacceptable. To Wheeler it seemed clear that the laws of general relativity must fail at the hole's center and be replaced by new laws of *quantum gravity,* and these new laws must halt the crunch. Perhaps, Wheeler speculated, building on views he had expounded in Brussels, the new laws would convert the imploding matter into radiation that quantum mechanically "tunnels" its way out of the hole and escapes into interstellar space. To test this speculation would require understanding in depth the marriage of quantum mechanics and general relativity. Therein lay the beauty of the speculation. It was a testbed to assist in discovering the new laws of quantum gravity.

As Wheeler's student in the early 1960s, I thought that his speculation of matter being converted into radiation at the singularity and then tunneling its way out of the hole was outrageous. How could Wheeler believe such a thing? The new laws of quantum gravity would surely be important in the singularity at the hole's center, as Wheeler asserted. But not near the critical circumference. The critical circumference was in the